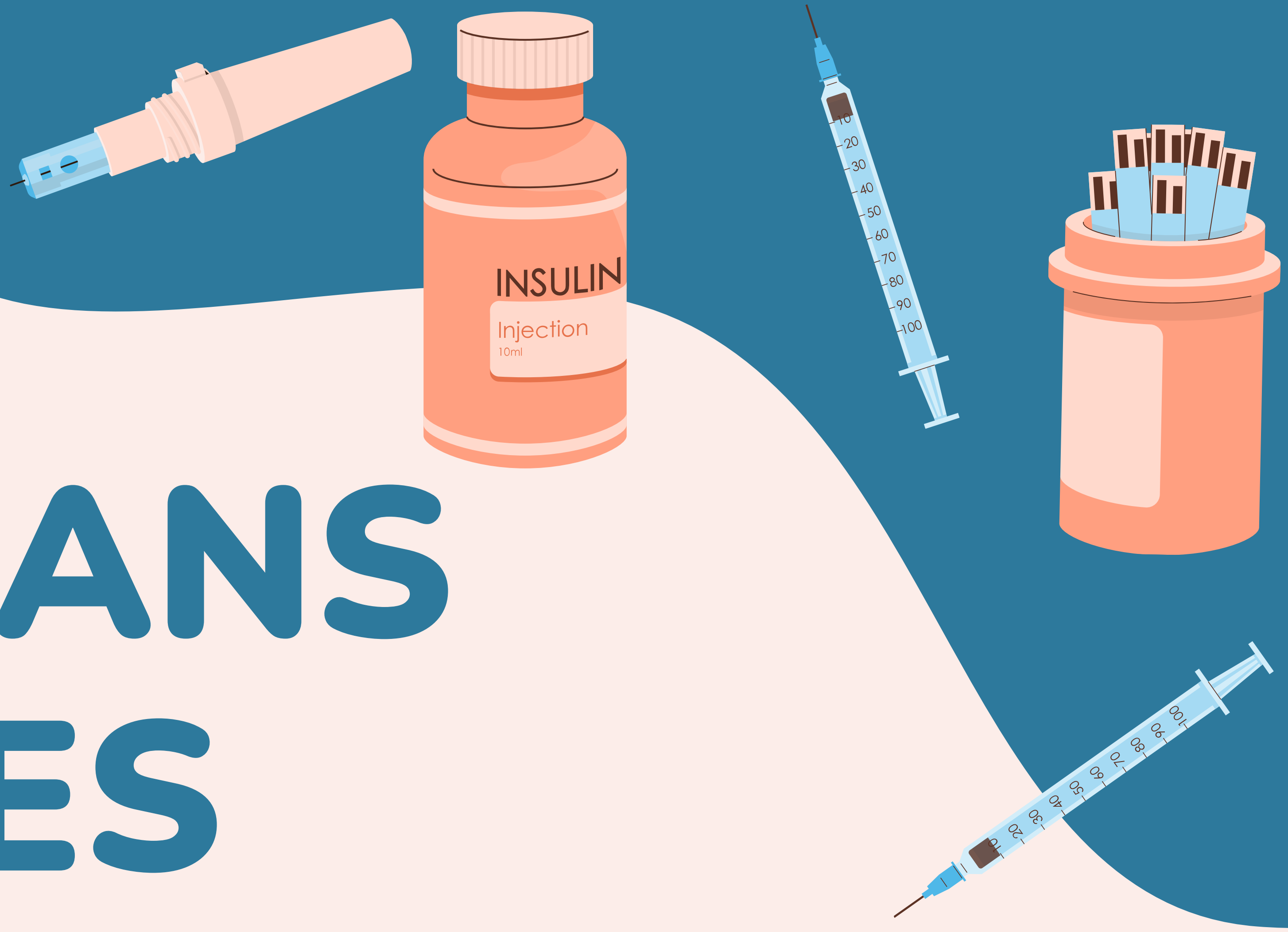

PIMA INDIANS DIABETES

GlucoMetrics Health

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AGENDA

- Company introduction
- Public health challenge
- Dataset Overview
- Risk Factor Analysis & Discussion
- Recommendations

ABOUT OUR TEAM

Data Science Consulting Team at GlucoMetrics Health

Analyze diabetes risk factor data from a landmark epidemiological study of the Pima Indian population

TASK

Client: Public Health Department

- Developing Diabetes Prevention Program

Data Given: Pima Indians Diabetes Database

- Longitudinal, epidemiological Study
    - Data collection began in the mid-1960s

Objectives

- Identify which risk factors best predict diabetes
- See which factors are modifiable
- Make appropriate recommendations

Pima Indians Diabetes Database

- **Source**: National Institute of Diabetes and Digestive and Kidney Diseases
- **Observations**: 768 female Pima Indian Heritage patients (21+) living near Phoenix, Arizona, USA
- **Variables**: Several independent variables, one dependent variable (outcome)
- **Unique Feature**: Diabetes Pedigree Function (DPF) is unique feature of dataset

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

PUBLIC HEALTH CHALLENGE

Diabetes is a chronic disease that occurs either when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces (World Health Organization, 2024).

Pima Indians have the highest recorded prevalence and incidence of non-insulin-dependent diabetes mellitus globally
(Knowler et al., 1978; King and Rewers, 1991)

The prevalence of diabetes for Native Americans is higher for women than for men
(rates vary among tribes)

DEEP DIVE INTO THE DATASET

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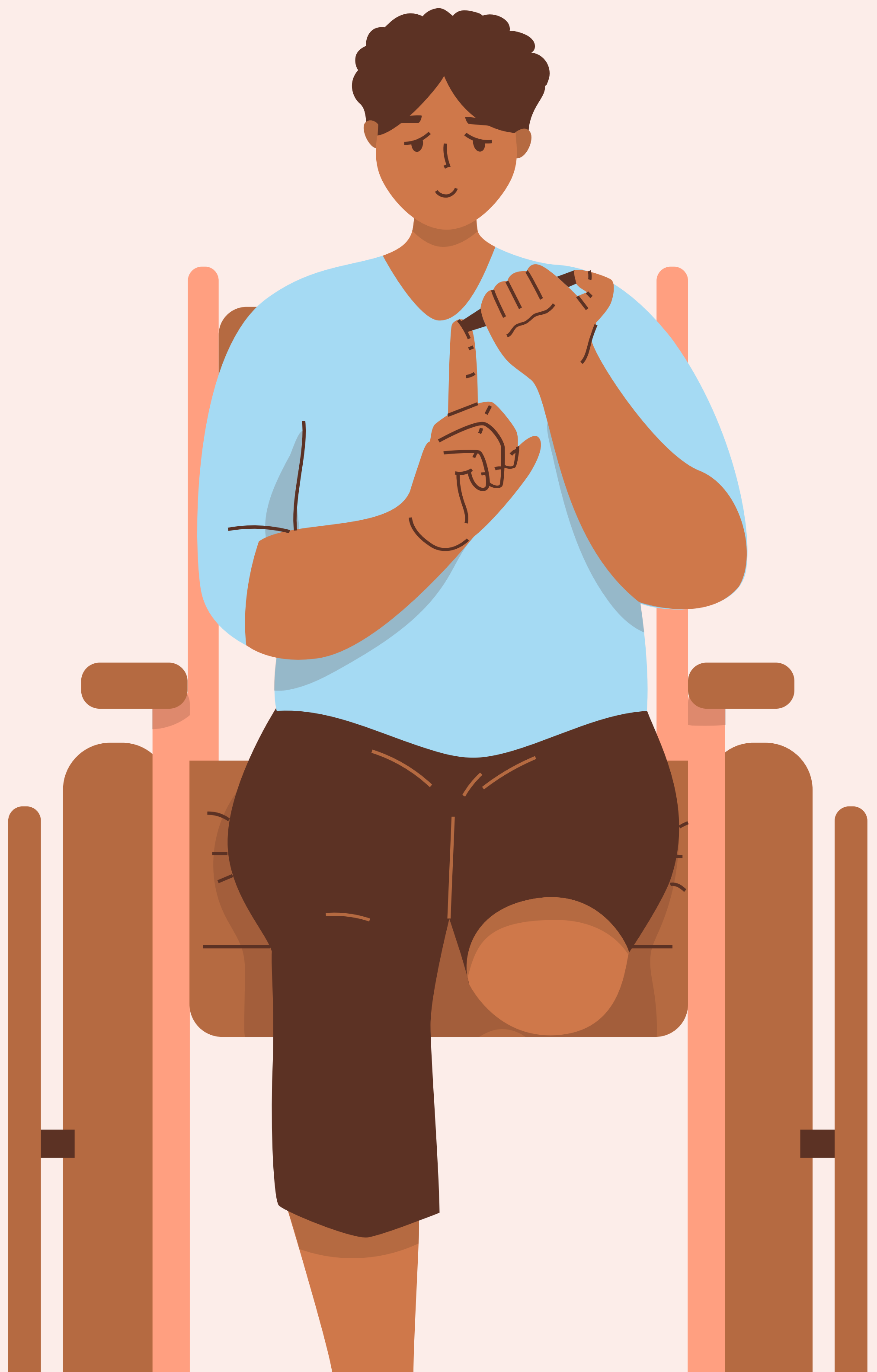


Variables and Data Types

- **Pregnancies:** Number of times pregnant
- **Glucose:** Plasma glucose concentration after a 2-hour oral glucose tolerance test (mg/dL)
- **BloodPressure:** Diastolic blood pressure (mmHg)
- **SkinThickness:** Tricep skinfold thicknesss (mm)
- **Insulin:** 2-hour serum insulin (μU/mL)
- **BMI:** Body mass index (weight in kg / height in m^2)
- **DiabetesPedigreeFunction:** A numeric score used to estimate an individual's genetic risk of developing type 2 diabetes based on their family history (Zhang, 2025)
- **Age:** Age of the patient in years
- **Outcome:** 0 = No diabetes, 1 = Diabetes

10 - Point Inspection

- **Shape:** 768 observations and 9 features (6,912 total data points)
- **Columns:** 9 total columns
- **Data Type:** All variables were int64, except BMI and DPF (float64)
- **First Look:** Noticed values that look like placeholders for missing data
- **Last Look:** Data ended cleanly and remains consistent

```
Data Types:
Pregnancies                   int64
Glucose                       int64
BloodPressure                 int64
SkinThickness                 int64
Insulin                       int64
BMI                         float64
DiabetesPedigreeFunction    float64
Age                           int64
Outcome                       int64
dtype: object
```

10 Point Inspection

- **Memory Usage:** 0.06 MB, very small dataset
- **Missing Values:** No missing values according to .isnull( )
- **Duplicates:** No duplicates
- **Descriptive Statistics:** Identified ranges and min values of 0 representing biologically impossible measurements
- **Unique Values:** All features had high cardinality, except outcome (binary).

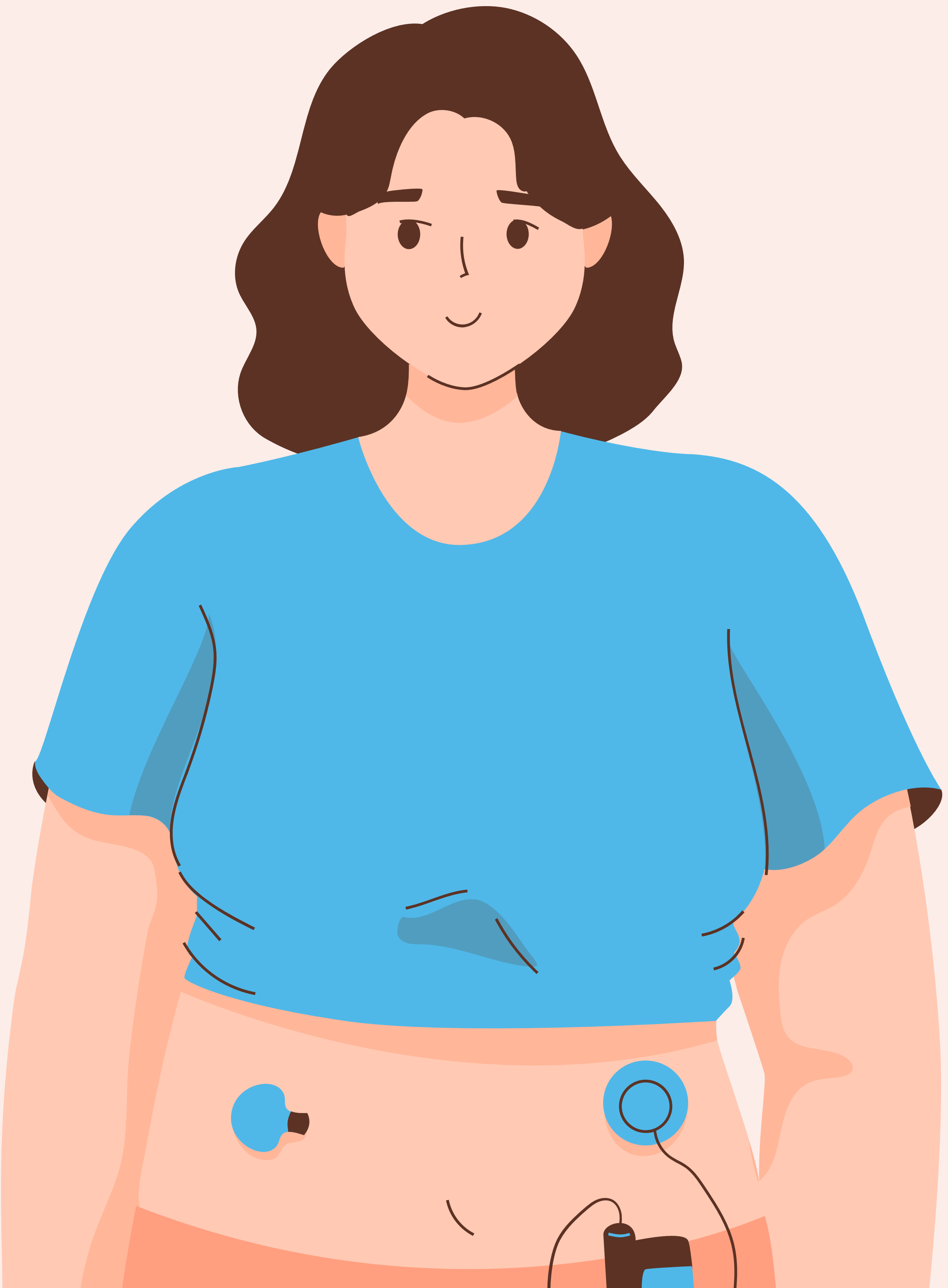
```
Percentage of Missing Values by Column:
Pregnancies                 0.00%
Glucose                     0.00%
BloodPressure               0.00%
SkinThickness               0.00%
Insulin                     0.00%
BMI                         0.00%
DiabetesPedigreeFunction    0.00%
Age                         0.00%
Outcome                     0.00%
dtype: object

Total Missing Values in Dataset: 0
✅ According to .isnull(), there are no missing values in this dataset.
```

Challenges with Dataset

This dataset uses 0 to represent missing values
Standard .isnull( ) checks would not detect these as missing

Handled by:

- Identifying which columns where 0 is biologically impossible
    - Glucose, BloodPressure, SkinThickness, Insulin, BMI
- Replacing impossible zeroes with NaN (placeholder for missing data)
- Analyzing only non-zero values



RISK FACTOR ANALYSIS

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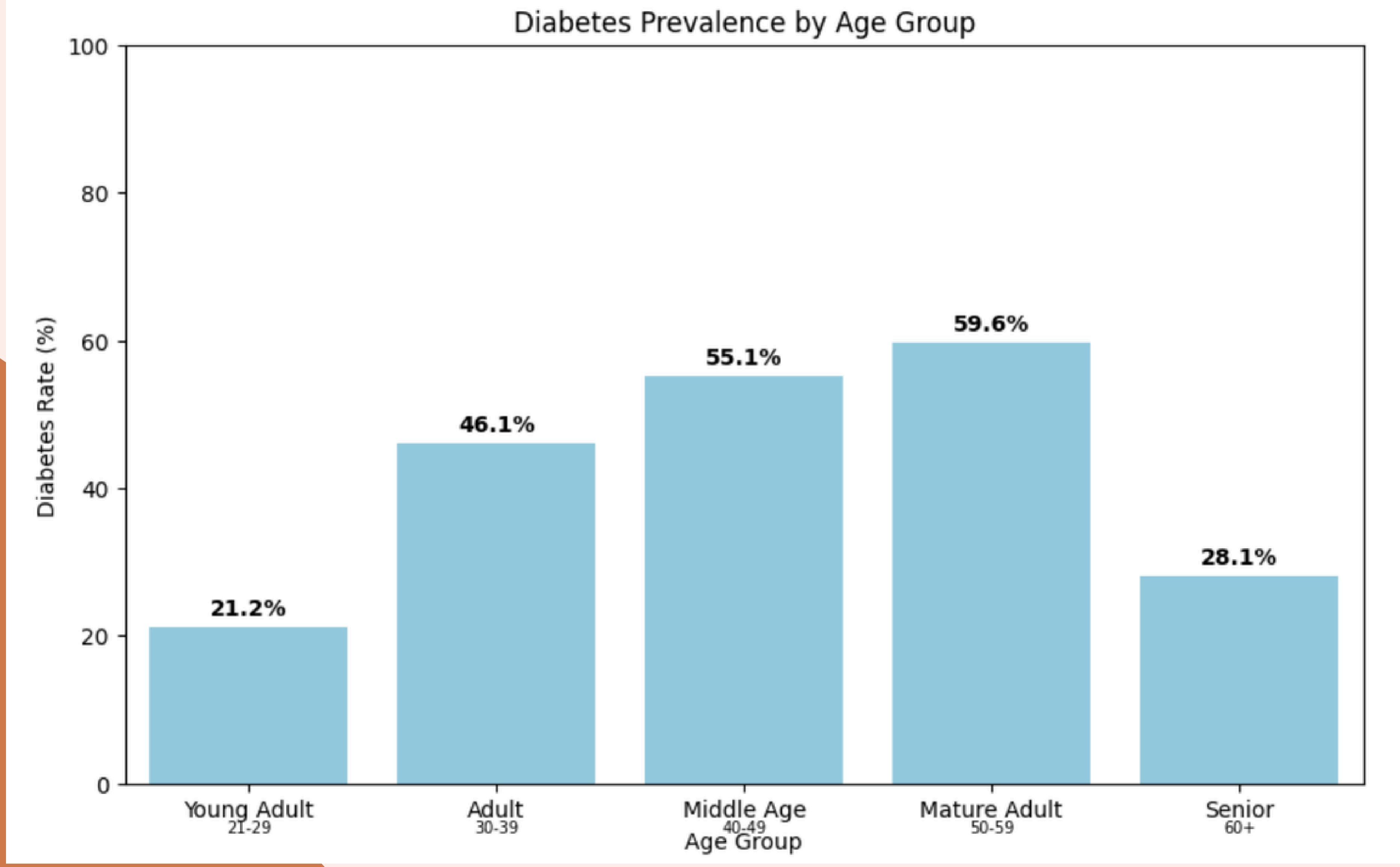
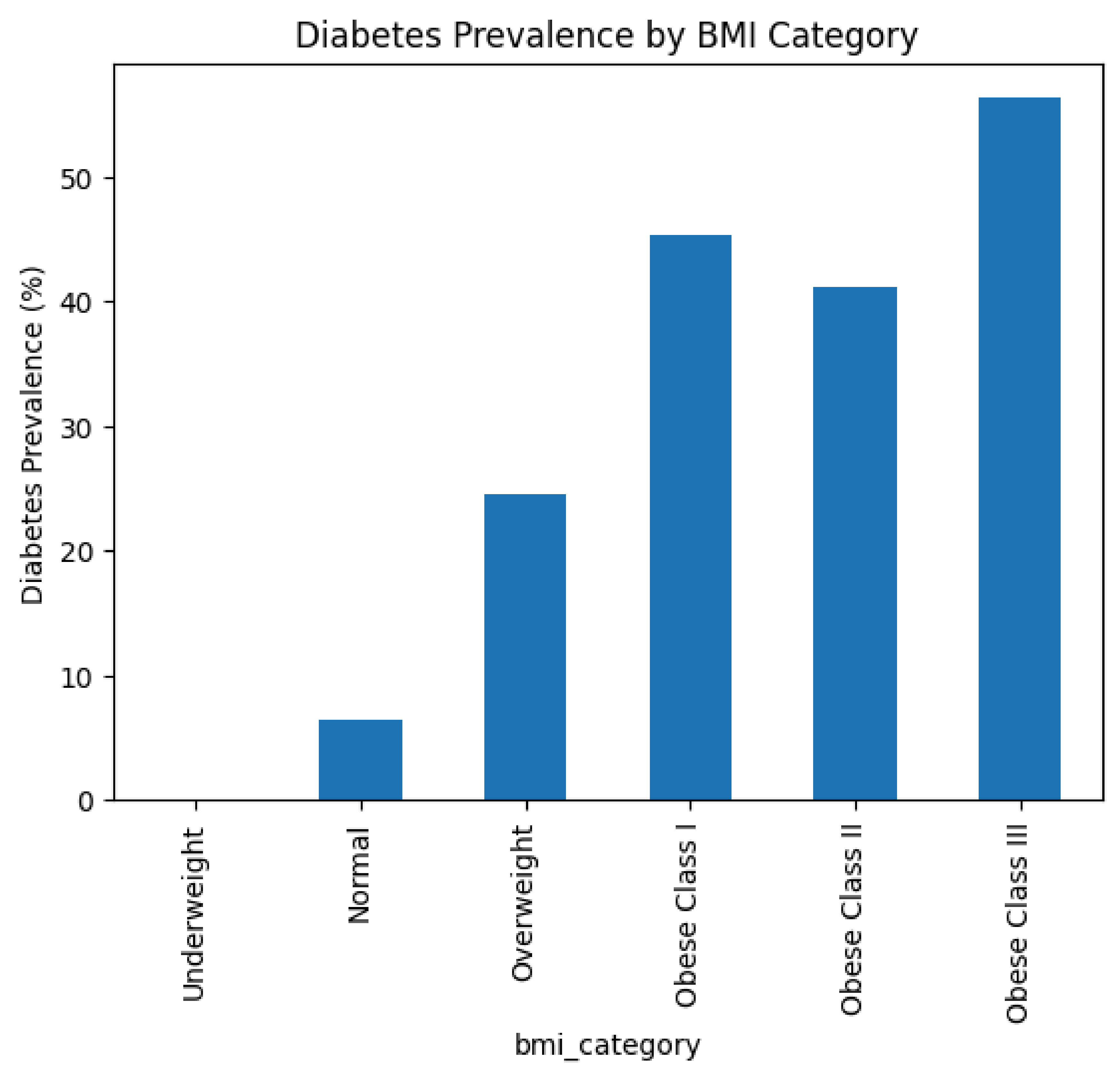
Age



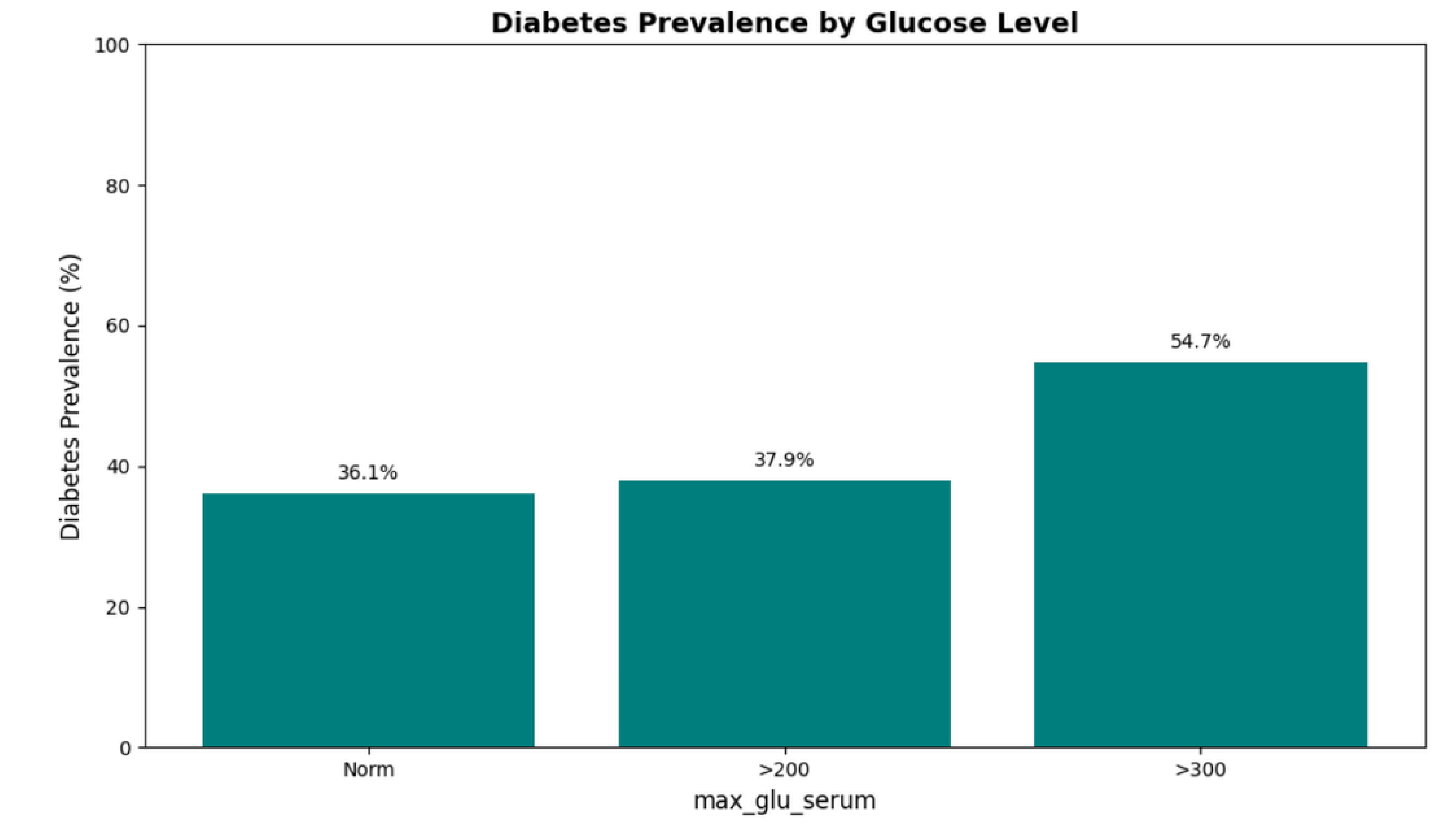
- Sharp increase from Young Adults (21.21%) to Adults (46.06%)
- Prevalence continues to rise with age before declining in Seniors
- Overall pattern supports research that **diabetes risk increases with age**
- Higher prevalence with age may reflect increased insulin resistance and accumulated health risk factors.

BMI



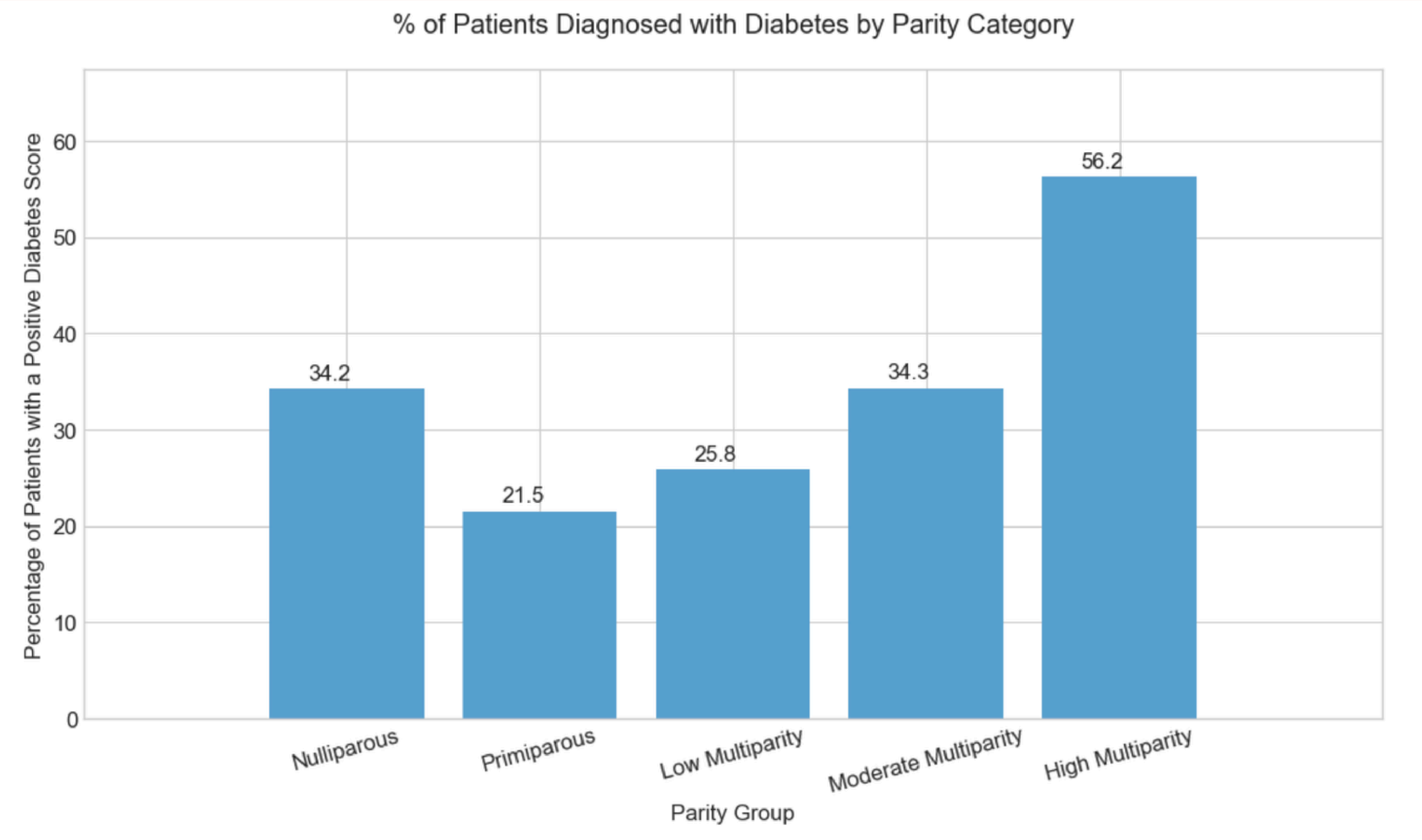
- Normal weight individuals have very low prevalence (~6%).
- Obese Class III is the highest (~56%).
- A substantial increase in risk beginning at BMI ≥ 25.
- Diabetes prevalence increases steadily as BMI category increases.
- The evidence suggests that **BMI is a significant risk factor for diabetes** in this population.

Glucose



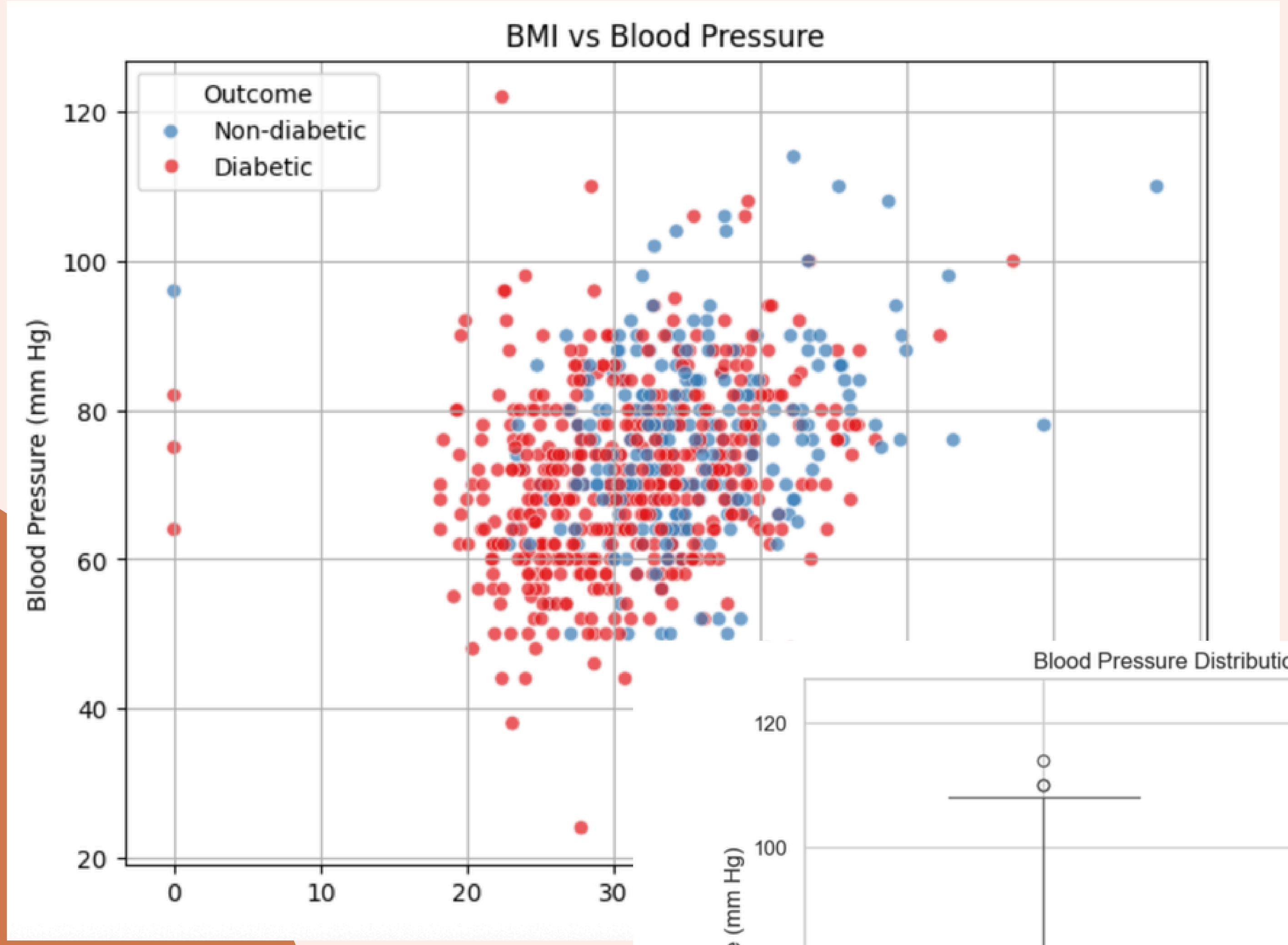
- Normal glucose individuals have moderate prevalence (~36%).
- >300 is the highest (~55%).
- Diabetes prevalence increases as glucose level increases.
- As the insulin levels rise, glucose rise is also correlated.
- The evidence suggests that **glucose level is a significant risk factor for diabetes** in this population.

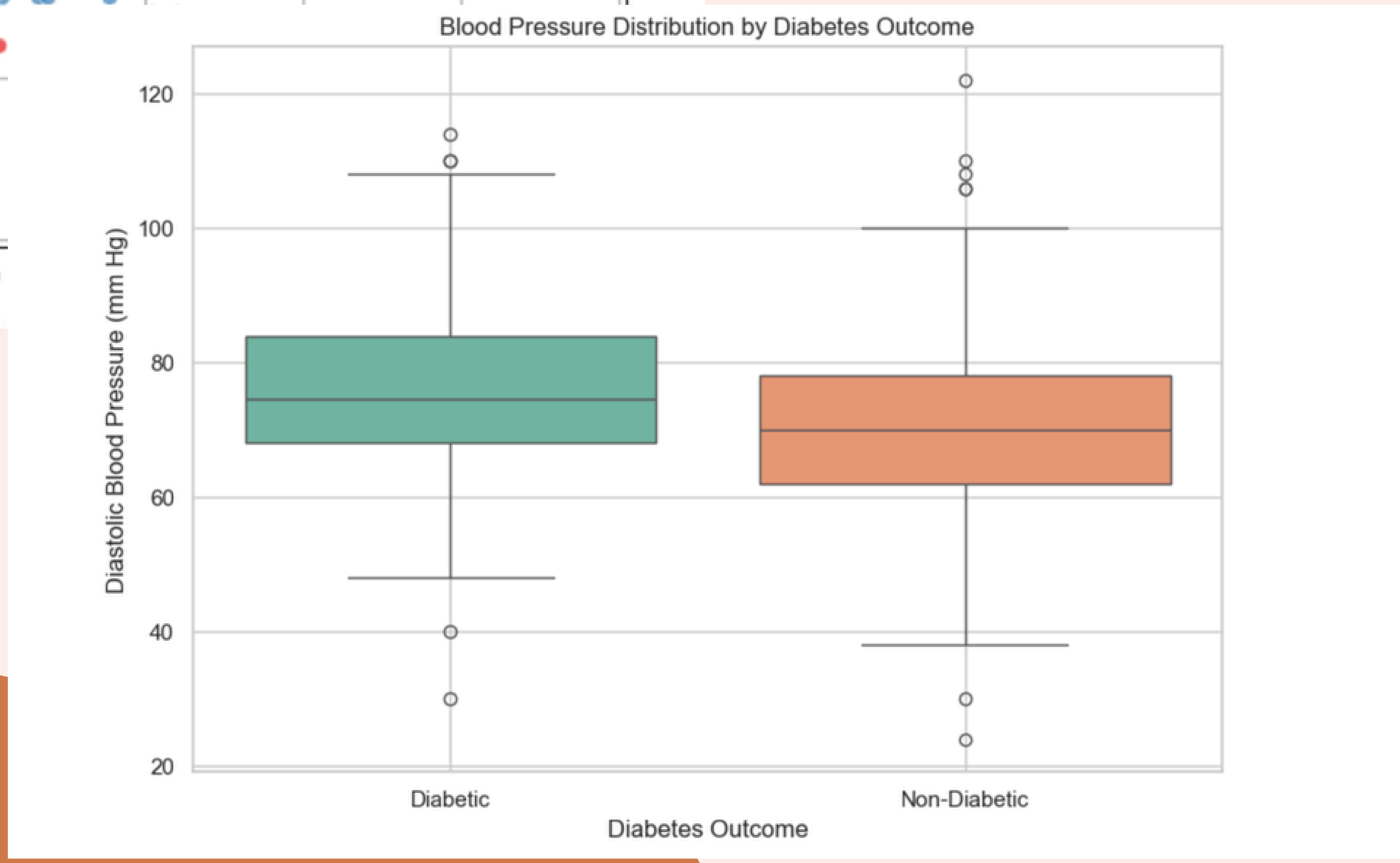
Pregnancy



- Individuals diagnosed with diabetes increases to **56.2%** for people with 7+ pregnancies.
- After 1 pregnancy, the possibility of the individual being diagnosed with diabetes gradually increases, suggesting a correlation of the two values.

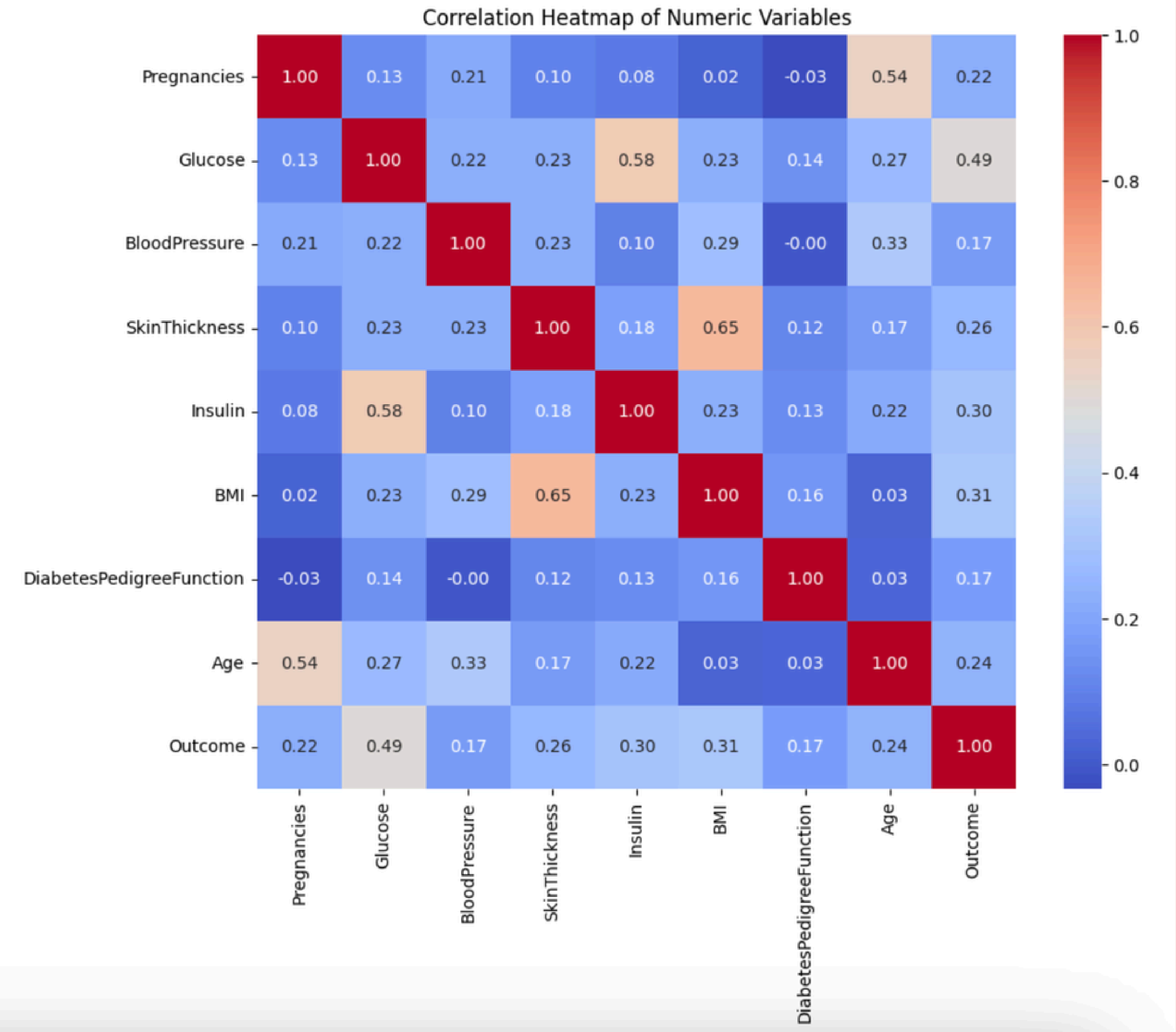
Blood Pressure





- BMI vs Blood Pressure
- Blood Pressure vs Diabetes Outcome
  - Individuals with higher blood pressure are more likely to have diabetes
- 46.8% of individuals with higher blood pressure, had diabetes vs 31.7%

WHICH RISK FACTORS ARE THE STRONGEST?



	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
Pregnancies	1.00	0.13	0.21	0.10	0.08	0.02	-0.03	0.54	0.22
Glucose	0.13	1.00	0.22	0.23	0.58	0.23	0.14	0.27	0.49
BloodPressure	0.21	0.22	1.00	0.23	0.10	0.29	-0.00	0.33	0.17
SkinThickness	0.10	0.23	0.23	1.00	0.18	0.65	0.12	0.17	0.26
Insulin	0.08	0.58	0.10	0.18	1.00	0.23	0.13	0.22	0.30
BMI	0.02	0.23	0.29	0.65	0.23	1.00	0.16	0.03	0.31
DiabetesPedigreeFunction	-0.03	0.14	-0.00	0.12	0.13	0.16	1.00	0.03	0.17
Age	0.54	0.27	0.33	0.17	0.22	0.03	0.03	1.00	0.24
Outcome	0.22	0.49	0.17	0.26	0.30	0.31	0.17	0.24	1.00

Strongest Risk Factors
(Correlation with Diabetes Outcome):

- Glucose (r = 0.49)
- BMI (r = 0.31)
- Insulin (r = 0.30)

Glucose is more strongly associated with diabetes outcome than other measured risk factors.

HOW DO RISK FACTORS INTERACT?

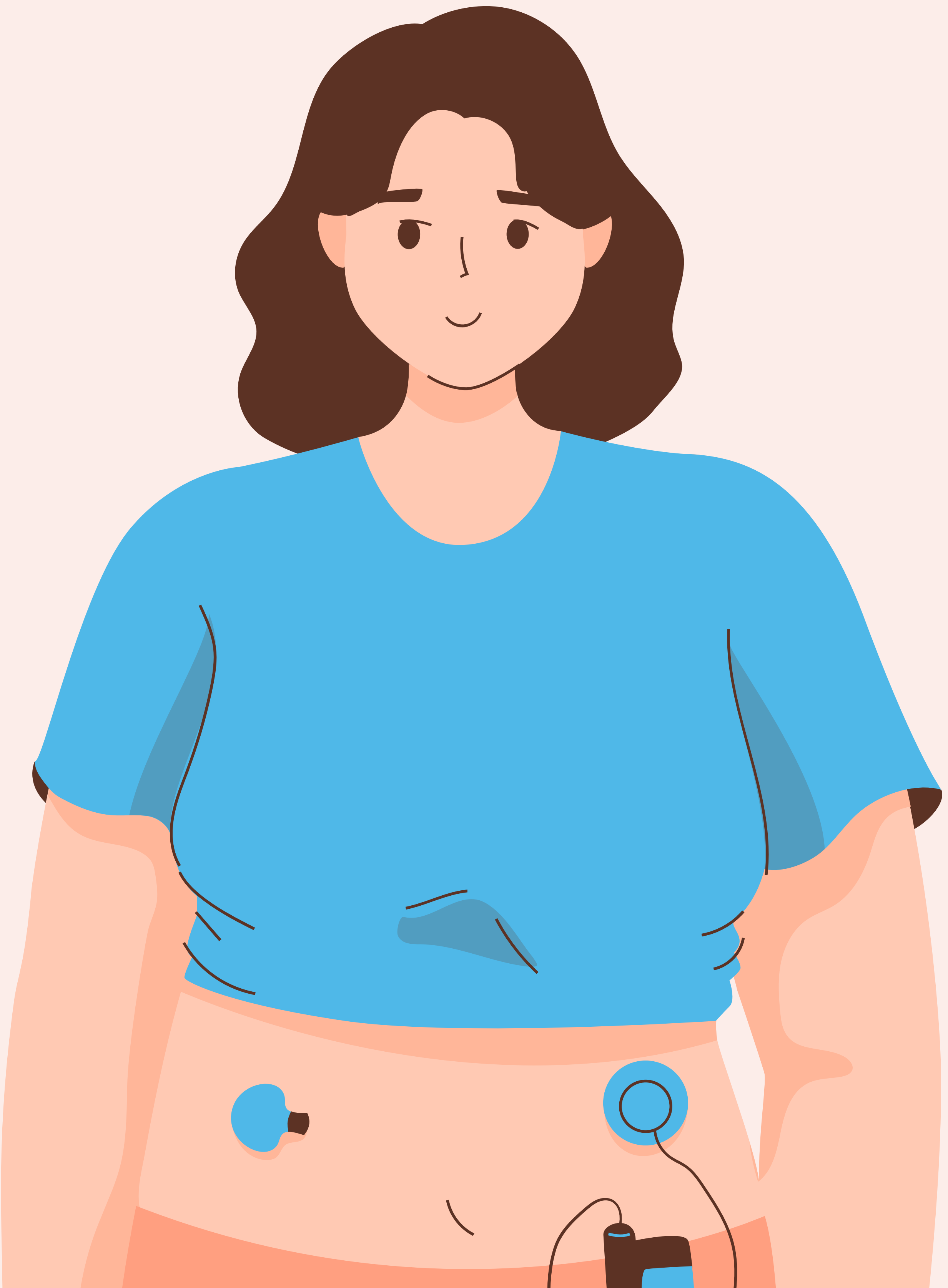
- **BMI ↔ Insulin ↔ Glucose:** Higher BMI increases insulin resistance, raising blood glucose levels.
- **Age ↔ Insulin:** Increasing age reduces insulin sensitivity and adds metabolic strain.
- **Pregnancies ↔ Glucose:** More pregnancies (especially with gestational diabetes) increase later glucose dysregulation risk.
- **Metabolic Syndrome:** High BP, high BMI, insulin resistance, and abnormal glucose often cluster together.

(Mayo Clinic, 2025)

WHICH RISK FACTORS ARE MODIFIABLE?

BMI: diet, exercise, weight management
Glucose levels: diet, exercise, medication
Blood pressure: diet, exercise, medication
Insulin levels: diet, exercise, weight management, medication
Age and Pregnancies are non-modifiable

(American Heart Association, 2024)



RECOMMENDATIONS

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WHAT SHOULD THE PREVENTION PROGRAM PRIORITIZE?

- **Primary Focus:** Glucose control and BMI reduction
  - Strongest correlations + highly modifiable
- **Secondary Focus:** Improve insulin sensitivity through lifestyle intervention
- **Target High-Risk Groups:** Older adults and women with multiple pregnancies
- **Overall Strategy:** Emphasize sustainable diet, physical activity, and early screening

PROPOSED DIABETES PREVENTION PROGRAM

- **Lifestyle Intervention:** Nutrition education, physical activity, and weight management programs to reduce BMI and improve metabolic health
- **Screening & Monitoring:** Regular glucose, BMI, and blood pressure screenings for early detection and timely clinical intervention
- **Targeted Outreach:** Focus on high-risk groups (older adults, women with multiple pregnancies or history of gestational diabetes)
- **Ongoing Support:** Health coaching, behavioral counseling, and digital tracking tools to promote long-term habit change

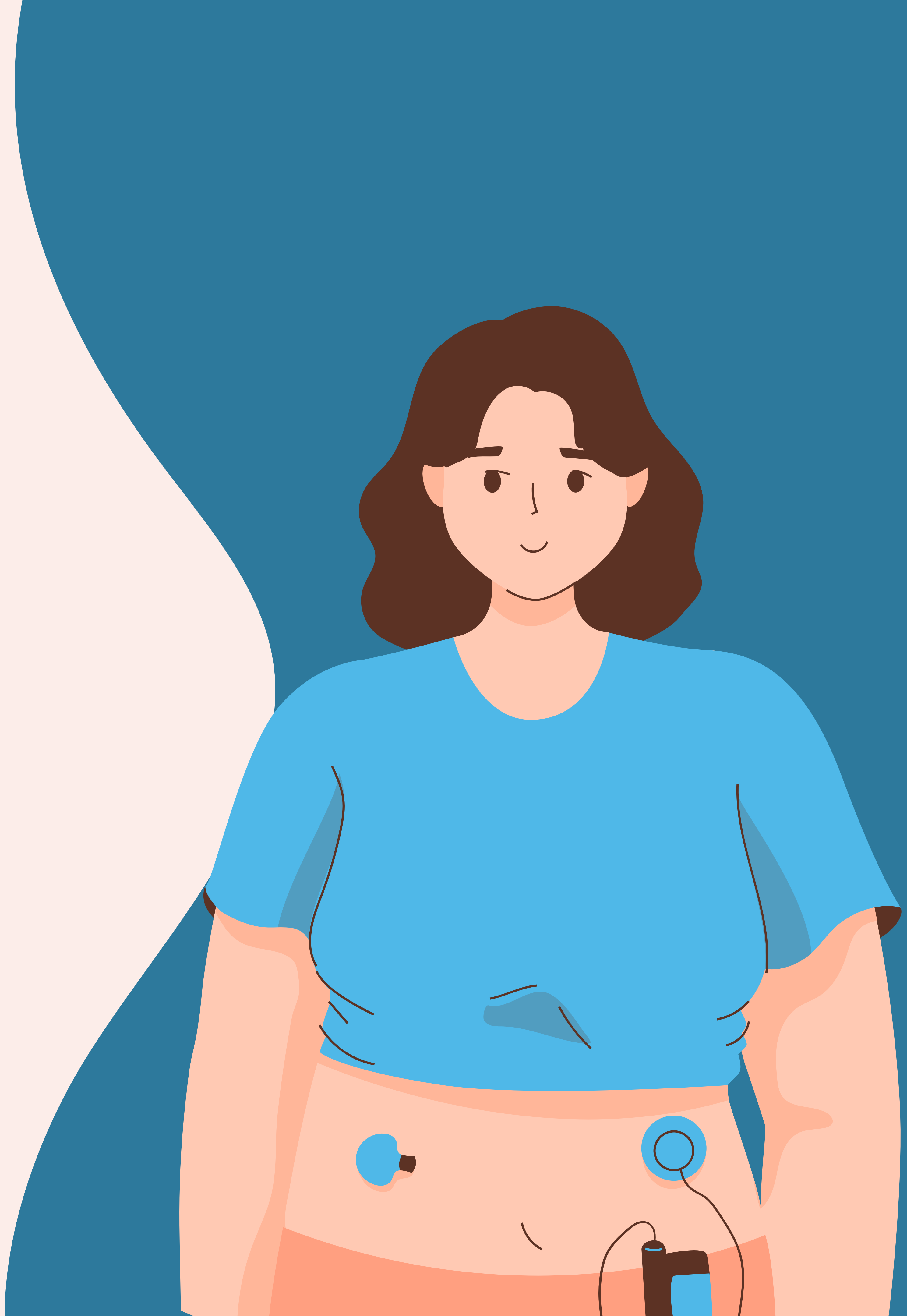
WHAT ARE THE LIMITATIONS?

- **Missing Risk Factors:**
  - No data about diet, exercise, medication, or socioeconomic factors
- **Small sample size & Limited Representation:**
  - Only 768 records of female Pima Indians aged ≥21; findings may not generalize to males or other populations
  - Fewer diabetes cases than non-diabetes cases, limiting statistical power and model performance without adjustment
- **Binary Outcome Only**
  - No diabetes types or severity data, restricting deeper analysis
- <u>Current analysis does not capture interactions between risk factors or predictive modeling capacity.</u>

CONCLUSION

- Glucose and BMI were the strongest risk factors associated with diabetes outcome.
- Modifiable factors (BMI, glucose, insulin, blood pressure) present key opportunities for prevention.
- Age and pregnancy history help identify high-risk groups for targeted screening.

- **Future Work**
  - Develop predictive models to assess interactions among risk factors.
  - Evaluate how well risk factors predict diabetes beyond simple correlations.



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THANK YOU!

Do you have any questions, comments, or suggestions?